EV Customer Prioritization

Nu Quynh Chau Nguyen | Individual project

I propose a decision support tool for an illustrative EV dealership with limited outreach capacity. A sales or customer relationship manager would use a ranked list to decide which customer profiles to review first.

Question and intended use

Which K customer profiles are most likely to have Will_Buy_EV = Yes in the approved dataset? For example, if staff can contact 50 people, the output would be a reviewable list of 50 profiles with relevant driving and charging information. A staff member decides whether and how to contact them.

Proposed method and simple comparison

The topic is supervised binary classification. The main method is Logistic Regression, following the course classification example [7]. Encode the target as No = 0 and Yes = 1. The model first calculates a weighted score z, then applies the sigmoid function [8]:

$$z = b_0 + b_1x_1 + ... + b_mx_m$$

$$p(\text{Yes} \mid x) = \text{sigmoid}(z) = 1 / (1 + \exp(-z))$$

Here x represents the prepared predictors. Standardize numeric fields and one-hot encode categories using training data only. Learn the weights by minimizing binary cross-entropy (log loss), with regularization; the initial library implementation uses L-BFGS [8]. This fits a Yes/No target and produces probabilities for ranking.

Compare with a Dummy baseline that uses the training Yes-label rate for every profile. Also report the majority-class rule that predicts No for everyone. Rank the Logistic Regression probabilities to select exactly K profiles; this is separate from using a 0.5 classification cutoff.

Success measure

Precision@K is the share of Yes labels among the K selected profiles. Lift@K divides that share by the evaluation set's Yes-label rate. Later evaluation will use K = 10% of each evaluation batch as the primary comparison. K = 50 illustrates staff capacity. A useful ranking improves on the baseline (Lift > 1) on separate evaluation data; this is not a guaranteed sales target.

Scope

The professor-approved Kaggle competition contains synthetic profiles [1]. Scores estimate its supplied label, not confirmed purchases or extra sales caused by outreach. No partner dealership, real campaign, customer names, or phone numbers are claimed.

Repository: https://github.com/Nuquynhchaunguyen/ev-customer-prioritization

Analytics topics considered

Each example connects a topic to a realistic business decision.

Topic	Realistic example	Decision supported
Supervised learning	A dealership learns from labeled profiles to estimate an EV purchase label.	Prioritize a fixed number of profiles for staff review.
Unsupervised learning	A retailer groups customers by shopping and service needs without outcome labels.	Design suitable support or outreach for different groups.
Dimensionality reduction	An operations team summarizes many correlated vehicle or charging measurements.	Create a simpler view while checking what information is lost.
Natural language processing	A service team categorizes written EV support requests by topic.	Route charging, maintenance, and financing questions to the right team.
Semantic or sentiment analysis	A brand team examines permitted public EV reviews for opinions about charging.	Identify concerns that need clearer information or further research.
Digital channel analytics	A dealership compares permitted campaign engagement summaries across channels.	Choose which messages or posting times to test next.

Selected topic

I selected supervised learning because the approved data have an explicit Yes/No target and structured customer characteristics. Logistic Regression links those predictors to a probability score that can support a capacity-limited ranking [7,8].

Why the other topics are not the main approach

The dataset has no review text, support messages, campaign channel histories, or observed contact outcomes. NLP, sentiment analysis, or channel analytics would need another suitable source. Unsupervised grouping alone does not estimate the supplied target, and dimensionality reduction is not needed to begin with thirteen predictors.

Business connection

Customer-response research provides a relevant example of using classification and lift for targeting [6]. For EV consultation, daily driving and available charging options are practical questions [2,3]. The proposed output helps a manager prepare a review list; its usefulness must be evaluated before real outreach.

Supervised customer prioritization

From binary response modeling to an EV outreach decision

Learn from labeled examples, estimate a probability, and rank profiles for a human decision. Logistic Regression is the project's main method; the milestones below show its development and a related industry use.



Berkson [4]
Logistic function applied to biological response modeling.

Cox [5]
Binary response regression relates outcome probabilities to explanatory variables.

Moro et al [6]
Bank telemarketing research compares classifiers and assesses targeting with lift.

This project
Synthetic EV labels support a proposed shortlist for dealership review.

Practical use and EV industry connection



Campaign research
A Portuguese banking study used response prediction and lift to examine targeting [6]. This is an industry example, not evidence of EV sales impact.



EV consultation
Daily travel and home, workplace, or public charging options affect the questions a prospective EV buyer should consider [2,3].

Potential value and costs



Capacity
Review a shortlist sized to the team's available contact time.



Human judgment
Staff confirm customer needs and suitability.



Resources
Data checks, implementation, and validation require time.

Connection to the selected dataset

The competition supplies 668,665 labeled profiles and thirteen predictors [1]. The proposed model learns the supplied Yes/No label and ranks profiles for review. Any business benefit remains to be tested; the synthetic source does not demonstrate real customer response or guaranteed savings.

Figure 1. Topic and industry infographic. Timeline: [4-6]. EV context: [2,3]. Dataset: [1]. All icons and arrows are original vector drawings. Full references appear on page 5.

Dataset profile and first findings

Approved Kaggle source | Playground Series Season 6 Episode 9 [1]

One row is one synthetic customer profile. The labeled file contains id, thirteen predictors, and Will_Buy_EV. The competition test file has no target, so it cannot measure predictive accuracy or ranking quality.

File	Rows	Columns	Purpose
train.csv	668,665	15	Labeled development data
test.csv	286,571	14	Unlabeled scoring data
sample_submission.csv	286,571	2	Kaggle output template

Key fields

Six numeric predictors: age, annual income (USD), daily commute (km), cars owned, and charging-station counts near home and work. Seven categorical predictors: gender, city type, current car type, home charging possible, subsidy available, range anxiety, and environmental concern. The concern values 1-5 have undocumented survey anchors and stay nominal categories. Exclude id from predictors.

Quality and preparation decisions

Checks found zero missing cells, blank text, duplicate IDs/full rows, and checked domain violations in train and test. IQR flags identified 3,678 incomes, 41 commute distances, and 13,489 car counts in labeled data; flags alone do not prove errors. Retain valid values: zero source rows or predictor values were removed or changed. [Project evidence: data-quality and IQR audits.]

Class balance

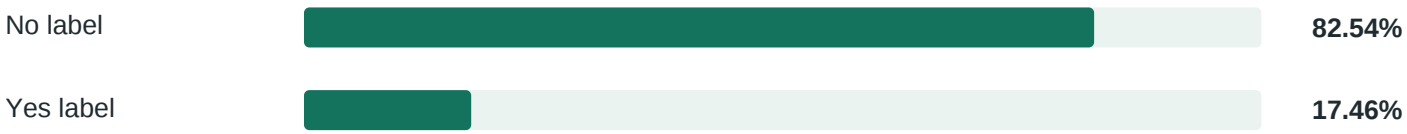


Figure 2. No: 551,886 profiles (82.54%); Yes: 116,779 (17.46%). Source: target_summary.csv. The uneven labels mean accuracy alone can look good while missing the group the dealership wants to find.

Simple baseline and interpretation

Predicting No for everyone gives 551,886 / 668,665 = 82.54% accuracy but 0% Yes recall. This descriptive baseline establishes why later evaluation should examine the top-K list. Random selection has an expected Yes share near 17.46% and expected Lift near 1; an actual random shortlist can vary.

Access: use ev_project/download_data.py with your own Kaggle credentials and applicable competition access. Follow source licensing and competition rules. Credentials and full raw CSVs stay outside GitHub. Country, income definition, station radius, and purchase timeframe are not supplied. Supporting tables and notebook 01 record the checks.

Feasibility and next steps

The labeled target, accessible files, and initial checks support proceeding with the proposed method.

Next: review this proposal with the professor; define training, validation, and holdout roles; fit preparation steps on training data only; compare Logistic Regression with the Dummy baseline; report Precision@K and Lift@K on separate evaluation data. Review subgroup results and predictor choices before any real use.

Still unknown: performance for actual dealership customers, real-world probability calibration, and the effect of outreach. No contact outcomes exist in this source. P2 will develop the technical analysis and working handoff using feedback from P1.

References

[1] Kaggle. Predicting Electric Vehicle Purchases. Playground Series Season 6 Episode 9. Approved competition dataset. Source.

[2] U.S. Department of Energy. Alternative Fuels Data Center. Electric Vehicles for Consumers. Driving and charging considerations. Source.

[3] International Energy Agency (2026). Global EV Outlook 2026. Electric vehicle charging. Source.

[4] Berkson, J. (1944). Application of the Logistic Function to Bio-Assay. Journal of the American Statistical Association, 39(227), 357-365. Source.

[5] Cox, D. R. (1958). The Regression Analysis of Binary Sequences. Journal of the Royal Statistical Society Series B, 20(2), 215-232. Source.

[6] Moro, S., Cortez, P., and Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. Decision Support Systems, 62, 22-31. doi:10.1016/j.dss.2014.03.001. Source.

[7] Solis, S. MSBA265 Project 02 Classification. Course assignment instructions: logistic score and sigmoid classification. Source.

[8] Scikit-learn 1.9.1 documentation. Logistic Regression: probability model, log loss, regularization, and optimization. Source.